

Yousef Kakhki

Senior Backend Engineer & Tech Lead

Tehran, Iran | Open to remote work and international relocation (EU Blue Card / Netherlands HSM eligible)

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1 SUMMARY

Senior Backend Engineer and Technical Lead with 10 years of experience shipping distributed systems in Fintech and Media – including a WebRTC/HLS streaming platform serving 5,000+ concurrent users and a production-ready crypto-to-fiat payment gateway pending regulatory authorization. Comfortable at every layer:

NATS JetStream signaling, PostgreSQL ledger design, Linux kernel tuning, and Yocto OTA pipelines for 5,000 IoT devices. Led engineering teams of up to 12. M.Sc. in Computer Science (System Design), Amirkabir University of Technology (Tehran Polytechnic), GPA 3.8/4.0. IELTS 7.5 (C1). EU Blue Card eligible; available immediately.

2 WORK EXPERIENCE

September 2025 – Present Solutions Architect (Project Contract)

HonarAmoozesh – Remote

- Architected a production-grade hybrid streaming platform combining WebRTC (LiveKit) and HLS delivery, serving 5,000+ concurrent users while reducing cloud infrastructure costs by 78% compared to the previous single-protocol setup.
- Engineered a real-time signaling layer achieving sub-100 ms latency using NATS JetStream, enabling seamless user role promotion and dynamic room management for live interactive sessions.
- Deployed scalable LiveKit Selective Forwarding Unit (SFU) clusters behind HAProxy load balancing with Prometheus monitoring and automated alerting, achieving 99.9% uptime during peak usage periods.
- Designed intelligent network-condition fallback mechanisms that transparently switched users between WebRTC and HLS streams, eliminating buffering events and improving perceived quality of experience.
- Tech stack: LiveKit, WebRTC, HLS, NATS JetStream, HAProxy, Prometheus, Docker, Linux.

October 2023 – June 2025 Head of Software & AI Development

Capitalino (CITEX Spin-off) – Tehran, Iran (Fintech / Crypto exchange platform)

- Architected and deployed a crypto-to-fiat payment gateway built for high-volume operation: idempotent transaction handling, Kafka-based event reconciliation, automated dispute resolution, and multi-provider fiat routing to eliminate single points of failure. Production-ready; regulatory authorization pending at project close.
- Architected and deployed a Digital Investment Management Dashboard with real-time portfolio analytics, risk assessment algorithms, automated rebalancing logic, and blockchain API integration for crypto asset tracking across 5+ exchanges.
- Led the AI Hologram Project from concept to production – computer vision pipeline (pose estimation) feeding WebSocket-controlled holographic projection, end-to-end latency under 80ms; won Best Booth at ITEX 2024 among 200+ companies.
- Built and managed a 4-person engineering team via self-hosted GitLab CI/CD; maintained multiple production releases per week throughout the engagement.
- Oversaw server security hardening and disaster recovery protocols; hybrid cloud infrastructure with centralised logging (ELK Stack) and monitoring (Prometheus/Grafana).
- Tech stack: Node.js, TypeScript, Apache Kafka, PostgreSQL, Redis, Docker Swarm, GitLab CI/CD, Prometheus, Grafana, ELK Stack, Python.

April 2021 – September 2023 Senior Backend Engineer

Avin Avisa – Tehran, Iran (High-frequency algorithmic trading, cryptocurrency markets)

- Architected a sub-100 ms peer-to-peer (P2P) order matching engine with price-time priority matching and idempotent ledger semantics, achieving throughput of 10,000+ requests per second.
- Implemented an in-memory order book with persistent snapshot recovery, reducing system restart downtime from minutes to under 5 seconds.
- Designed Atomicity, Consistency, Isolation, Durability (ACID)-compliant transactional ledgers using PostgreSQL with row-level locking to guarantee financial integrity across all trading operations.
- Integrated Web3.js for on-chain verification and blockchain wallet management, supporting 3+ blockchain networks with zero double-spend incidents.
- Reduced average API response times by 60% through Redis caching strategies and targeted database query optimisation, directly improving trader experience under peak market conditions.
- Tech stack: Node.js, TypeScript, PostgreSQL, Redis, Web3.js, NATS, RabbitMQ, Docker, Linux.

September 2019 – March 2021 Systems Engineer – Embedded Linux

Batna – Tehran, Iran (IoT edge devices / embedded systems)

- Modified Linux kernel drivers and boot sequences using systemd parallelisation, reducing system initialisation time by 40% on ARM-based embedded hardware.
- Designed an atomic Over-the-Air (OTA) firmware update pipeline for 5,000+ IoT devices using dual A/B partitions, rollback mechanisms, and delta patching with secure HTTPS delivery, achieving zero bricked-device incidents over 18 months.
- Developed custom Yocto/Buildroot Linux distributions, integrating cross-compilation toolchains and CI/CD pipelines that cut firmware release cycles by 30%.
- Tech stack: Linux Kernel (C), systemd, Yocto, Buildroot, ARM, HTTPS/TLS, CI/CD.

April 2016 – August 2019 Android Systems Developer

Azma Data Structure – Mashhad, Iran (Enterprise Android applications)

- Developed native modules using Android NDK (C++) for GPS tracking, achieving sub-metre accuracy and 50% battery reduction through sensor fusion.
- Engineered offline-first state synchronisation with conflict resolution via Operational Transformation (OT) and background sync queues.
- Built scalable Android applications serving 10,000+ active users with biometric authentication, encrypted local storage (SQLCipher), and real-time FCM sync.
- Tech stack: Java, Android NDK (C++), SQLCipher, Firebase Cloud Messaging (FCM).

3 EDUCATION

Master of Science in Computer Science (System Design)

2018 – 2021

Amirkabir University of Technology (Tehran Polytechnic) – Tehran, Iran

Iran's #1 ranked university in Computer Science.

GPA: 3.8 / 4.0

- Thesis: "GreedyLearner" – A gamified algorithm-learning platform with adaptive learning paths and real-time code evaluation (Node.js, React, PostgreSQL, Docker).
- Blockchain Committee Lead: Designed a decentralised reward system using Hyperledger Sawtooth with custom transaction processors for academic credit tokenisation.
- Coursework: Advanced Algorithms, Distributed Systems, Cloud Computing, AI/ML.

Bachelor of Science in Computer Science

2012 – 2017

University of Bojnurd – Bojnurd, Iran

- Teaching Assistant for Data Structures, Advanced Algorithms, and Operating Systems; mentored 100+ undergraduate students across multiple semesters.
- Capstone: Distributed file storage system with Java RMI redundancy, achieving 99% uptime in stress testing.

4 TECHNICAL SKILLS

Architecture: Microservices, Event-Driven Systems, Domain-Driven Design (DDD), CQRS, API Gateway, Service Mesh, Distributed Systems, High-Availability, Load Balancing.

Backend: Node.js (Expert), TypeScript, Java, C++, Python, Go, RESTful APIs, GraphQL, WebSocket, gRPC, NATS JetStream, RabbitMQ, Apache Kafka.

Databases: PostgreSQL (Advanced), Redis (Caching/Pub-Sub), MongoDB, Elasticsearch, Sharding, Query Optimisation, Replication.

DevOps: Docker, Docker Swarm, Kubernetes (K8s), CI/CD (GitLab CI, Jenkins), Ansible, Terraform, Prometheus, Grafana, ELK Stack, Linux Administration.

Specialised: WebRTC (LiveKit SFU), HLS Streaming, Linux Kernel, Yocto/Buildroot (ARM), Hyperledger Sawtooth, Web3.js, AI/ML Operations.

5 CERTIFICATIONS

Introduction to TensorFlow for Artificial Intelligence – Coursera

Agile & Scrum Methodologies – Professional Certification

Hyperledger Sawtooth Architecture – Research-level study

References: Official Letter of Recommendation available from Capitalino CEO upon request.

6 LANGUAGES

- **Persian (Farsi):** Native proficiency
- **English:** Professional proficiency – IELTS 7.5 (C1)
- **French:** Professional working proficiency – B2 (Upper Intermediate)